EXPERIMENT-3

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1.COMMAND LINE INTERFACE:

PYTHON CODE:

```
import os
import sys

def rename_file(old_name, new_name):
    try:
        os.rename(old_name, new_name)
        print(f"File renamed from {old_name} to {new_name}")
    except FileNotFoundError:
        print(f"Error: {old_name} not found.")
    except Exception as e:
        print(f"An error occurred: {e}")


if __name__ == "__main__":
    if len(sys.argv) != 3:
        print("Usage: python rename_file_cli.py ")
else:
        rename_file(sys.argv[1], sys.argv[2])
```

OUTPUT:

```
D:\240701506>python rename_file_cli.py old.txt new.txt
File renamed from old.txt to new.txt
```

2.GRAPHICAL USER INTERFACE:

PYTHON CODE:

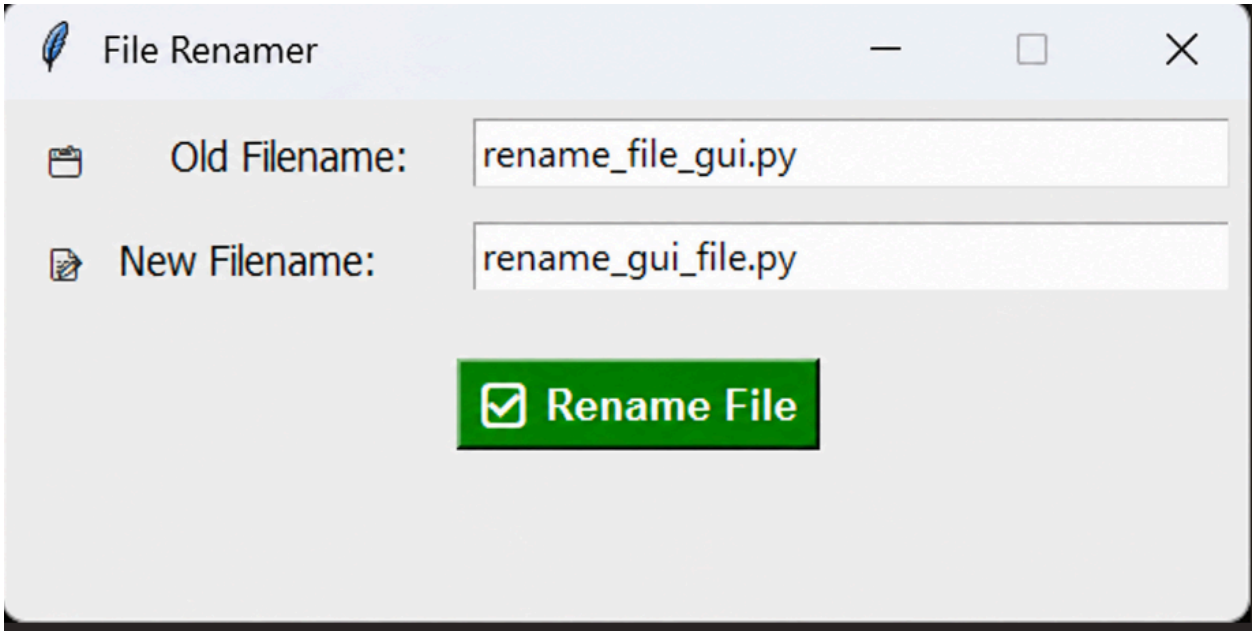
```
import tkinter as tk
from tkinter import messagebox
import os

# Function to rename the file
def rename_file():
    old_name = old_filename_entry.get().strip()
    new_name = new_filename_entry.get().strip()
# Check if input fields are empty
```

```
if not old_name or not new_name:
    messagebox.showwarning("Warning", "Please enter both filenames!")
    return

# Check if the old file exists
 if not os.path.exists(old_name):
 messagebox.showerror("Error", f"File '{old_name}' not found.")
 return

# Check if the new file already exists
if os.path.exists(new_name):
   overwrite = messagebox.askyesno("Warning", f"'{new_name}' already exists.
Overwrite?")
if not overwrite:
   return


try:
   os.rename(old_name, new_name)
   messagebox.showinfo("Success", f"File renamed from '{old_name}' to '{new_name}'")
except Exception as e:
    messagebox.showerror("Error", f"An error occurred: {e}")

# Create main window
root = tk.Tk()
root.title("File Renamer")
root.geometry("400x150")
root.resizable(False, False)

# Fixed window size
# Labels

tk.Label(root, text="🗂️ Old Filename:", font=("Arial", 10)).grid(row=0, column=0,
padx=10, pady=5, sticky="w")

tk.Label(root, text="📝 New Filename:", font=("Arial", 10)).grid(row=1, column=0,
padx=10, pady=5, sticky="w")

# Entry fields
old_filename_entry = tk.Entry(root, width=40)
```

```
old_filename_entry.grid(row=0, column=1, padx=10, pady=5)
new_filename_entry = tk.Entry(root, width=40)
new_filename_entry.grid(row=1, column=1, padx=10, pady=5)

# Styled Button
rename_button = tk.Button(root, text="✅ Rename File", bg="green", fg="white",
font=("Arial", 10, "bold"), command=rename_file)
rename_button.grid(row=2, column=0, columnspan=2, pady=10)

# Run the GUI event loop

root.mainloop()
```

OUTPUT:

```
D:\240701506>python rename_gui.py

D:\240701506>_
```



3.VOICE USER INTERFACE:
PYTHON CODE:

```
import speech_recognition as sr
```

```
import os
def rename_file_from_voice_command(command):
    try:
      words = command.lower().split(" ")
      if "rename" in words and "to" in words:
          rename_index = words.index("rename")
          to_index = words.index("to")

          # Extract old and new filenames
           old_name = words[rename_index + 1]
           new_name = words[to_index + 1]

         # Check if file exists
         if not os.path.exists(old_name):
             print(f"Error: File '{old_name}' not found.")
             return
         # Rename file
         os.rename(old_name, new_name)
         print(f"✅ File renamed from '{old_name}' to '{new_name}'")

         else:
                print("Invalid command format. Say: 'Rename oldfile.txt to newfile.txt'")
     except Exception as e:
         print(f"Error: {e}")
   def listen_for_command():
      recognizer = sr.Recognizer()
      mic = sr.Microphone()

      print("🎤 Listening for command to rename a file...")
      with mic as source:
          recognizer.adjust_for_ambient_noise(source)
          audio = recognizer.listen(source)
    try:
        command = recognizer.recognize_google(audio)
        print(f"🎙️ Command received: {command}")
        rename_file_from_voice_command(command)
     except sr.UnknownValueError:
         print("❌ Sorry, I couldn't understand the command.")
     except sr.RequestError as e:
      print(f"⚠️ Could not request results from Google Speech Recognition service; {e}")
```

```
if __name__ == "__main__":
    listen_for_command()
```

```
D:\240701506>python rename_vui.py
🎤 Listening for command to rename a file...
🎙 Command received: rename
Invalid command format. Say: 'Rename oldfile.txt to newfile.txt'

D:\240701506>python rename_vui.py
🎤 Listening for command to rename a file...
❌ Sorry, I couldn't understand the command.
```

```
[Running] python -u "d:\240701506\vui.py"
Listening for command to rename a file...
Command received: rename sample to Shark Shal
File renamed from sample to Shark
```

4.USER SATISFACTION COMPARISON:

PYTHON CODE:

```
def survey():
    print("Rate your satisfaction with the following interfaces (1-5):")


# Get user input for each interface
 try:
    cli_satisfaction = int(input("CLI (1-5): "))
    gui_satisfaction = int(input("GUI (1-5): "))
    vui_satisfaction = int(input("VUI (1-5): "))

   # Ensure valid ratings
    if not (1 <= cli_satisfaction <= 5 and 1 <= gui_satisfaction <= 5 and 1 <=
vui_satisfaction <= 5):
print("Please enter ratings between 1 and 5 only.")
return

# Display the ratings
print("\nYour satisfaction ratings:")
```

```
print(f"CLI: {cli_satisfaction}")
print(f"GUI: {gui_satisfaction}")
print(f"VUI: {vui_satisfaction}")

# Calculate the average satisfaction

avg_satisfaction = (cli_satisfaction + gui_satisfaction + vui_satisfaction) / 3
print(f"\nAverage Satisfaction Score: {avg_satisfaction:.2f}")

except ValueError:
    print("Invalid input! Please enter numbers between 1 and 5.")

 # Run the survey function
 if __name__ == "__main__":
    survey()
```

OUTPUT:

```
D:\240701506>python survey.py
Rate your satisfaction with the following interfaces (1-5):
CLI (1-5): 4
GUI (1-5): 5
VUI (1-5): 3

Your satisfaction ratings:
CLI: 4
GUI: 5
VUI: 3

Average Satisfaction Score: 4.00

D:\240701506>_
```